

Smart Metro Booking System

Project Planning Phase

Date	
Team ID	
Project Name	Smart Metro Ticket Booking System
Maximum Marks	5 Marks

1. Project Planning Phase

1. Purpose

This document sets out the project planning approach for digitizing metro ticket booking on ServiceNow. It turns the solution requirements and technology stack into a structured plan — covering scope, work breakdown, timeline, roles, and risk management — so implementation proceeds in a controlled, milestone-driven way.

2. Project Scope & Deliverables

The planning phase covers full delivery of the ServiceNow-based metro ticketing solution, from catalog configuration through go-live. Key deliverables include:

- A configured Service Catalog item with Source, Destination, Passenger Type, and Number of Tickets variables.
- Automated fare calculation logic, including concessions for Student/Senior Citizen categories.
- QR code generation and validation integrated with the booking and payment flow.
- Station Details Mapping covering routes, distances, and fare slabs.
- Passenger-facing UI pages across the Mobile App, Portal, and WhatsApp channels.
- A tested, production-ready solution deployed following UAT sign-off.

3. Work Breakdown Structure

The project is split into four major work streams — Requirements & Design, Development, Integration, and Testing & Deployment — each broken further into concrete work packages aligned to the project milestones.

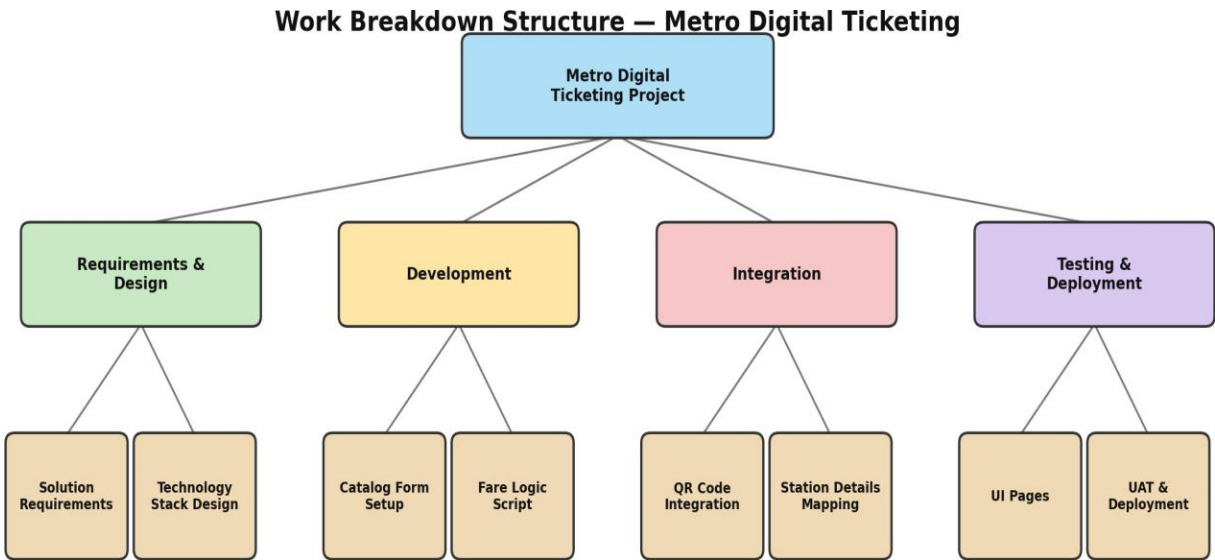


Figure — Work Breakdown Structure for Metro Digital Ticketing

4. Project Timeline & Milestones

Implementation follows six sequential milestones. The timeline below shows planned durations in project weeks, with the Fare Logic Script and Station Details Mapping running partly in parallel with Catalog Form Setup to shorten the overall schedule.

Project Timeline — Metro Digital Ticketing Implementation

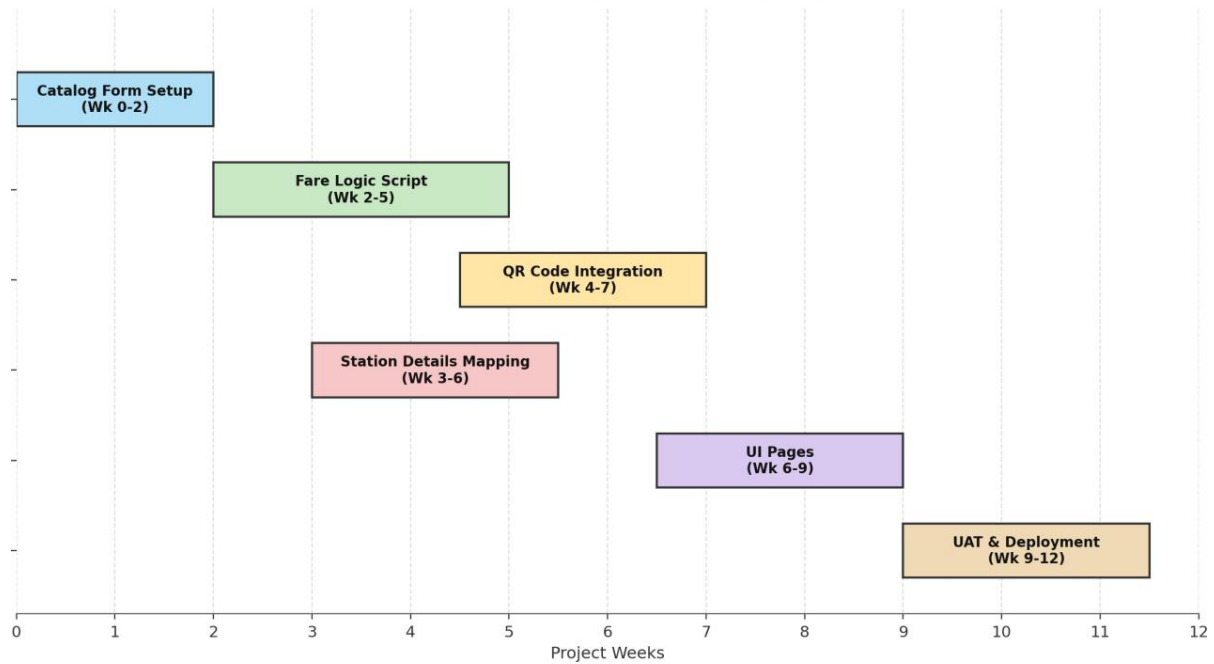


Figure — Indicative Project Timeline (in weeks)

4.1 Milestone Details

Milestone	Description	Owner
Catalog Form Setup	Configure the Service Catalog item with booking variables and validations.	IT Admin / ServiceNow Dev
Fare Logic Script	Build the fare calculation business rules and concession logic.	ServiceNow Developer
QR Code Integration	Integrate the QR generation API into the post-payment workflow.	Integration Developer
Station Details Mapping	Set up station master data, routes, and fare slabs.	Metro Operations Team
UI Pages	Build the passenger-facing booking and ticket display screens.	UI/UX Developer
UAT & Deployment	Run user acceptance testing and roll out to production.	QA Team / Project Manager

5. Roles & Responsibilities (RACI)

The RACI matrix below clarifies who is Responsible, Accountable, Consulted, and Informed for each major milestone.

Milestone	Passengers	Station Mgr	Ops Team	IT Admin
Catalog Form Setup	I	I	C	R/A
Fare Logic Script	I	I	C	R/A
QR Code Integration	I	I	I	R/A
Station Details Mapping	I	C	R/A	C
UI Pages	C	I	I	R/A
UAT & Deployment	C	C	A	R

R = Responsible, A = Accountable, C = Consulted, I = Informed.

6. Risk Management Plan

Risk	Impact	Mitigation
Delay in payment gateway API access/credentials	High	Start vendor onboarding in parallel with Catalog Form Setup.
Incomplete or inaccurate station master data	Medium	Validate station and fare data with the Operations Team before the Fare Logic Script begins.
QR code duplication or security gaps	High	Use signed/encrypted QR tokens and test tamper scenarios during UAT.
Low passenger adoption of digital channels	Medium	Pilot at select stations with in-person support before full rollout.
AFC gate hardware compatibility issues	Medium	Run early hardware integration testing during the QR Code Integration milestone.

7. Assumptions & Constraints

- The ServiceNow instance and required licenses (Catalog, Flow Designer, Integration Hub) are provisioned before the planning phase starts.
- Payment gateway and QR generation vendors are finalized and API access is available.
- Station Master Data is owned and maintained by the Metro Operations Team.
- UAT will involve representative passengers, Station Managers, and IT Admins.
- Go-live depends on successful UAT sign-off from all stakeholder groups.